

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-5005(3)-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Industrial Electronics

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the requirement of power supplies, which are used extensively in industrial application.
b) Write a short notes on the following:
 - i) Series voltage regulator
 - ii) Fly back converter
2. a) Draw and explain the block diagram of switched mode regulators.
b) The single phase full converter has RL load having $L = 6.5\text{mH}$, $R = 0.5\text{ ohm}$ and $E = 10\text{V}$ the input voltage is $V_s = 120\text{V}$ at (rms) 60Hz. Calculate:
 - i) Load current I_{LO} at $\omega t = \alpha = 60^\circ$
 - ii) The average thyristor current, I_A
 - iii) The average output current I_{dc}
3. a) Give the constructional features and principle of operation of Silicon Controlled Rectifiers (SCR).
b) What are the various methods to turn-on thyristor? Explain.

EC-5005(3)-CBGS

PTO

[2]

4.
 - a) What are the causes of damage to SCR. Also explain SCR overvoltage protection circuit.
 - b) Explain series and parallel operation of SCR and draw its characteristics.
5.
 - a) What are factors to be considered in selecting PLC and what are advantages of PLC.
 - b) Give functional block diagram of PLC.
6.
 - a) Explain with the help of ladder logic programming the process and control of PLC.
 - b) Explain working and operation of SMPS with the help of circuit diagram and waveform.
7.
 - a) Explain in brief the status and sequence of operations of PLC with the help of example.
 - b) What do you mean by ladder logic language explain in brief with the help of example.
8. Write short notes on any two of the following:
 - i) Power transistor
 - ii) Triac
 - iii) Function generator using OP-Amp.
 - iv) Simple process control applications of PLC
 - v) Factors to be considered in selecting PLC
